How representative are queries of information retrieval test collections?

Djoerd Hiemstra

https://www.cs.ru.nl/~hiemstra

@djoerd@idf.social

Open WebSearch

Are the queries we study a random sample of real queries?

Open Web Search

INFORMATION RETRIEVAL TEST COLLECTIONS

Empirical research in Web Search

The data:

- “Documents”: Lots of web pages
- “Topics”: A small sample of queries (say, 50)
  with a longer description of what the user wants to find
- “Relevance judgments”: For each query, the documents that are *relelevant* to that query

JOINT TASKS + REUSABLE COLLECTIONS

1) TREC (Text REtrieval Conference)
   - https://trec.nist.gov/
2) CLEF (Conference and Labs of the Evaluation Forum)
   - https://www.clef-initiative.eu/
3) NTCIR (NII Testbeds and Community for Information Research)
   - https://research.nii.ac.jp/ntcir/
4) FIRE (Forum for Information Retrieval Evaluation)
   - https://fire.irsi.org.in/

QUERIES (TOPICS) OF TEST COLLECTIONS

"Real" queries by real humans with an information need:

https://trec.nist.gov/data/t9.web.html (Hawking 2000)

```
<top>
<num> Number: 452
<title> do beavers live in salt water
<desc> Description:
Describe the normal habitat for beavers; note exceptions, if any.
<narr> Narrative:
Relevant documents describe the habitat range as well as
references to specific areas and bodies of water.
</top>
```

QUERIES (TOPICS) OF TEST COLLECTIONS

Mixed query stream with 75 home page, 75 named page, and 75 topic distillation queries

https://pages.nist.gov/trec-browser/trec13/web/overview/ (Craswell and Hawking, 2004)

```
Ireland consular information sheet
Citizen attitudes toward prairie dogs
JPL stardust comet wild
American music
Philadelphia streets
Togo embassy
Philippines
Baltimore
well water contamination
```

“Craswell & Hawking’s 2004 stratiefied sample”

Open WebSearch

Is Craswell & Hawking’s 2004 sample a random sample of real queries?

Open Web Search

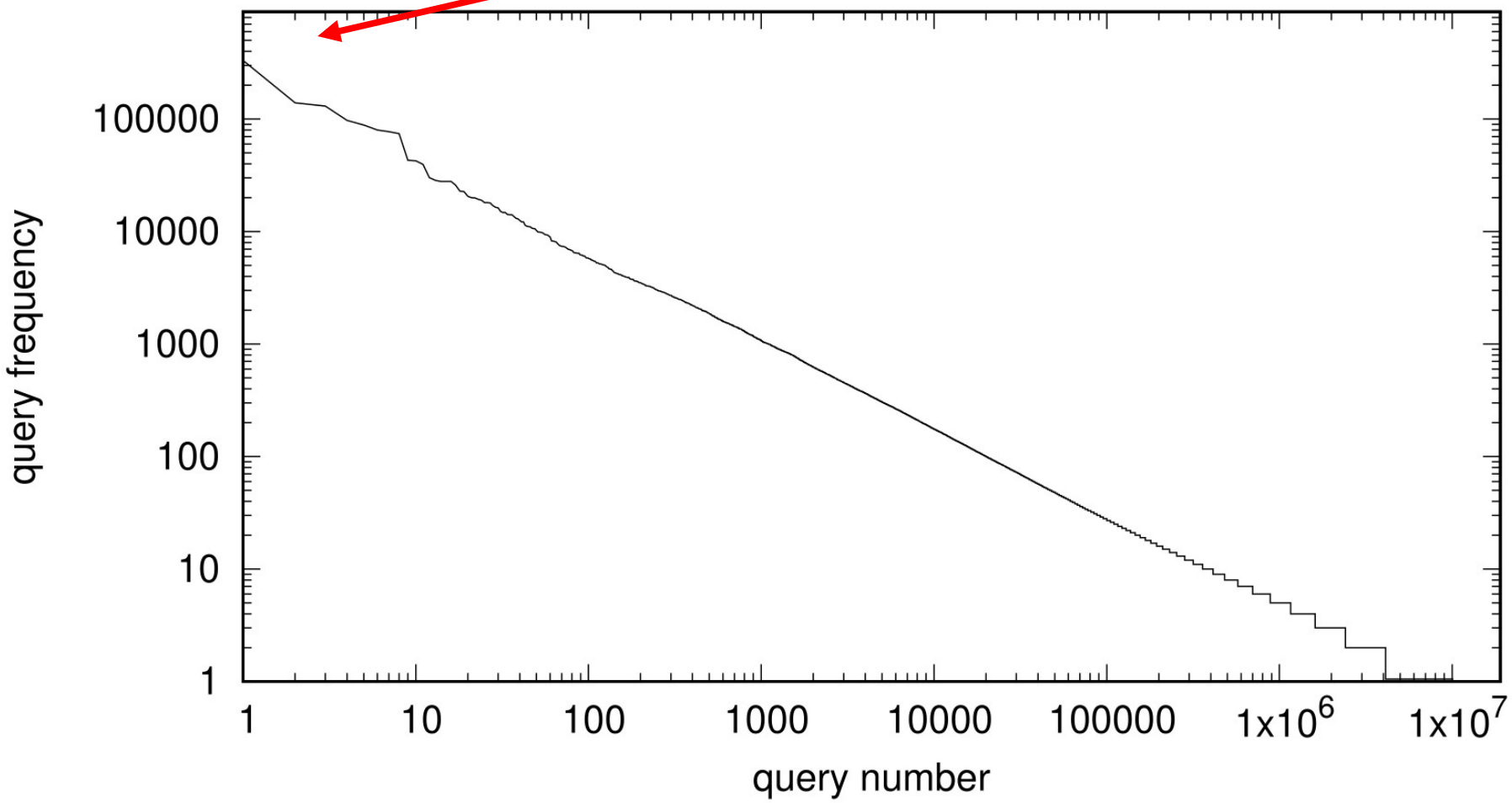
REAL QUERIES

Queries of America OnLine (AOL) Search engine (Pass et al, 2006)

- 1 March, 2006 - 31 May, 2006
- 36,389,567 lines of data
- 19,442,629 user click-through events
- 10,154,742 unique (normalized) queries

AOL LOG POWER LAW

most frequent query?



TOP SEARCHES IN AOL SEARCH (2006)

AOL query log (Pass et al, 2006)

1	google	332192
2	ebay	139207
3	yahoo	130538
4	yahoo.com	97518
5	mapquest	88279
6	google.com	79991
7	myspace.com	77211
8	myspace	74365

Skewed: “google” occurs almost 1% of the time!

IDEA

1) Divide the AOL log in 10 strata of 3,638,957 queries.
2) Assign a stratum to each query (out of 225) of Craswell & Hawking
3) Null hypothesis: Random sample
   - Then we expect about 20 queries in each stratum (ignore unseen queries?)
4) Chi-square test:
   - show that the probability of the sample is smaller than 0.01

ALSO

Probability to sample 225 queries from AOL

Not sampling “google”: 0.99

Not sampling “google” 225 times: $0.99^{225} = 0.10$

Skewed: “google” occurs almost 1% of the time!

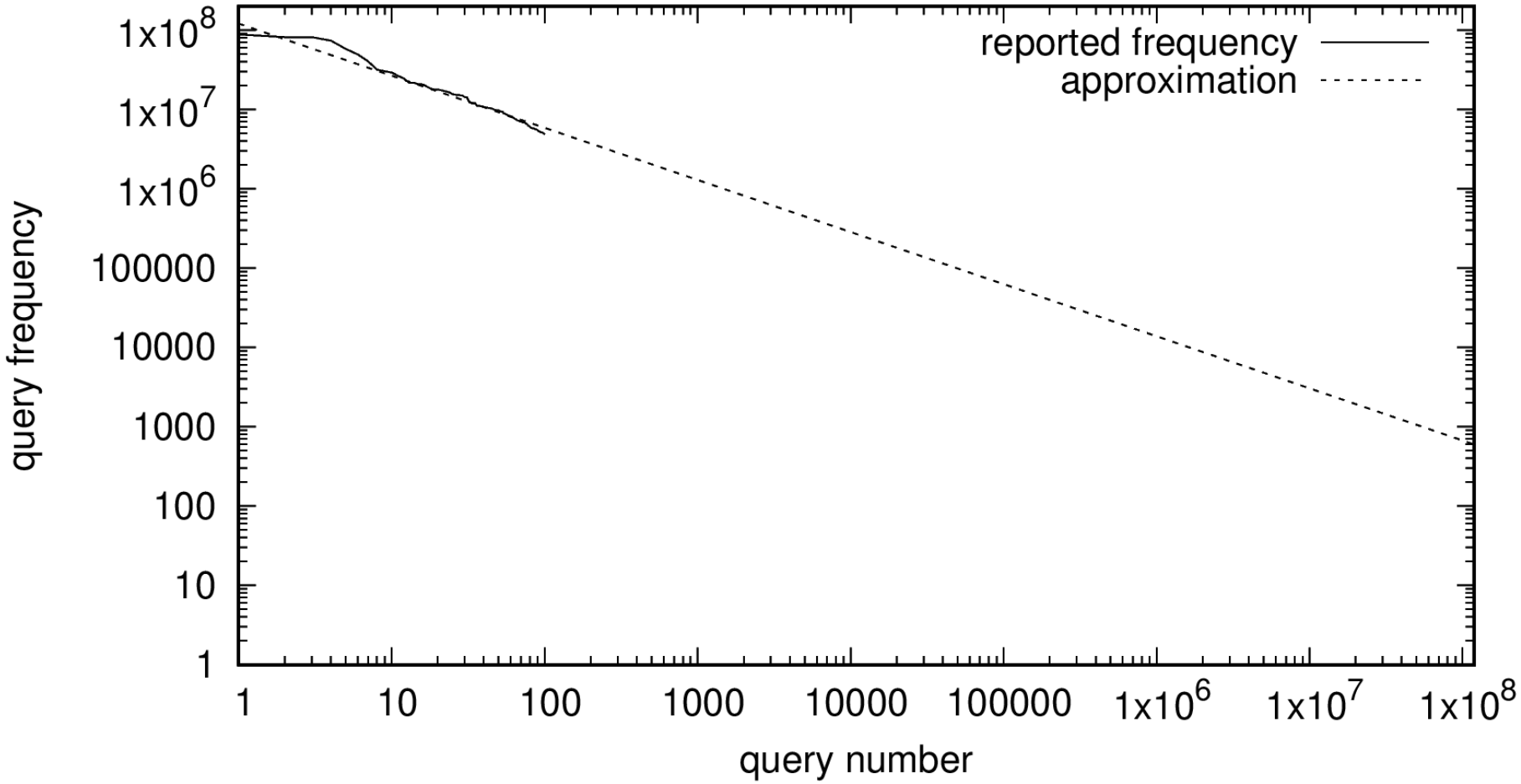
TOP WEB SEARCHES IN THE US (2026)

ahrefs.com (Search Engine Optimization)

https://ahrefs.com/blog/top-google-searches/

1	youtube	88300000
2	chatgpt	81340000
3	amazon	81170000
4	google	73230000
5	gmail	58120000
6	facebook	49090000
7	wordle	40370000
8	google translate	31600000

EXTRAPOLATION WITH LINEAR REGRESSION



SCRAPE STATISTICS FOR SINGLE QUERY

https://ahrefs.com/blog/marketing-tools/

ahrefs Dashboard Alerts Site explorer Content explorer Keywords explorer Rank tracker Site audit More

Google | best protein powder | United States

979 / 1,000 daily reports left 19,987 / 20,000 monthly SERP updates left

Overview
Ads history BETA

KEYWORD IDEAS
All keyword ideas
Phrase match
Having same terms
Also rank for
Search suggestions
Newly discovered
Questions

TRAFFIC SHARE
By domains
By pages

KEYWORDS LISTS

Overview: best protein powder

SERP & KD updated 21 hours ago. Update

+ Add to

Keyword difficulty

21
Medium

We estimate that you'll need backlinks from ~23 websites to rank in top 10 for this keyword

Search volume

51K

Not clicked 34%
Clicked 66%

RR 1.21
CPS 1.11

Since Mar 2017

Clicks 57K

CPC $1.10

Paid 17%
Organic 83%

Since Mar 2017

Global volume

78K

United States	51K	65%
United Kingdom	7.6K	9%
India	6.6K	8%
Canada	4.2K	5%
Australia	3.1K	3%
United Arab Em...	500	0%

Parent topic

best protein powder

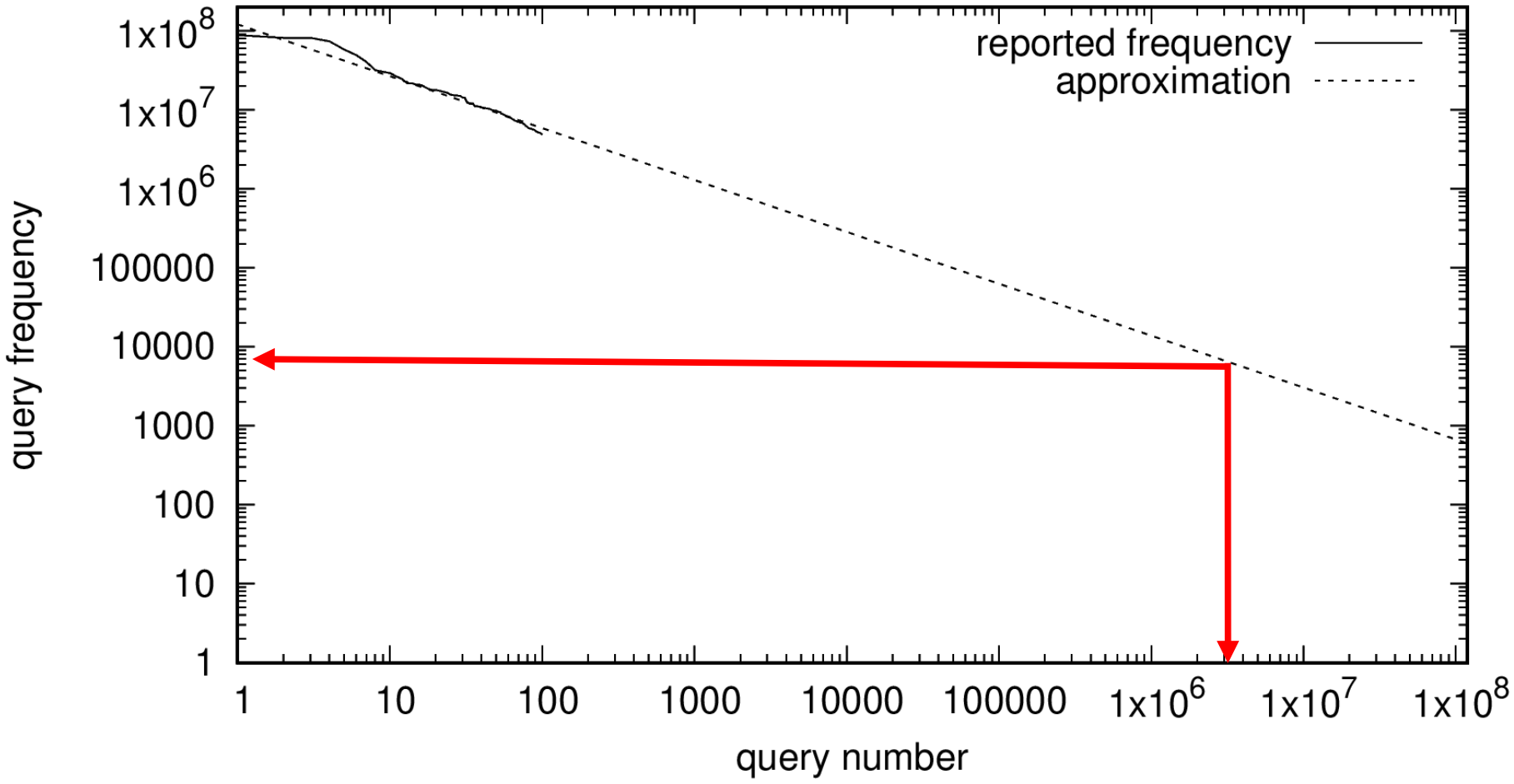
Search volume 51K

#1 result for parent topic

The 7 Best Types of Protein Powder - Healthline
https://www.healthline.com/nutrition/best-protein-powder

Total traffic 36K

EXTRAPOLATION WITH LINEAR REGRESSION



SCRAPE STATISTICS FOR SINGLE QUERY

https://ads.google.com/home/tools/keyword-planner/

Keyword plan

GO TO REPORTS TOOLS

Keyword ideas

Plan overview

Ad Groups

Keywords

Locations

Locations: United States Language: English Search networks: Google

Keyword ideas protein powder

DOWNLOAD KEYWORD IDEAS

Last 12 months
Nov 2017 - Oct 2018

Related terms to try: protein, optimum nutrition, protein bar, gold standard whey, whey protein isolate, discount supplements, casein protein

Found 2675 keyword ideas

FILTER COLUMNS

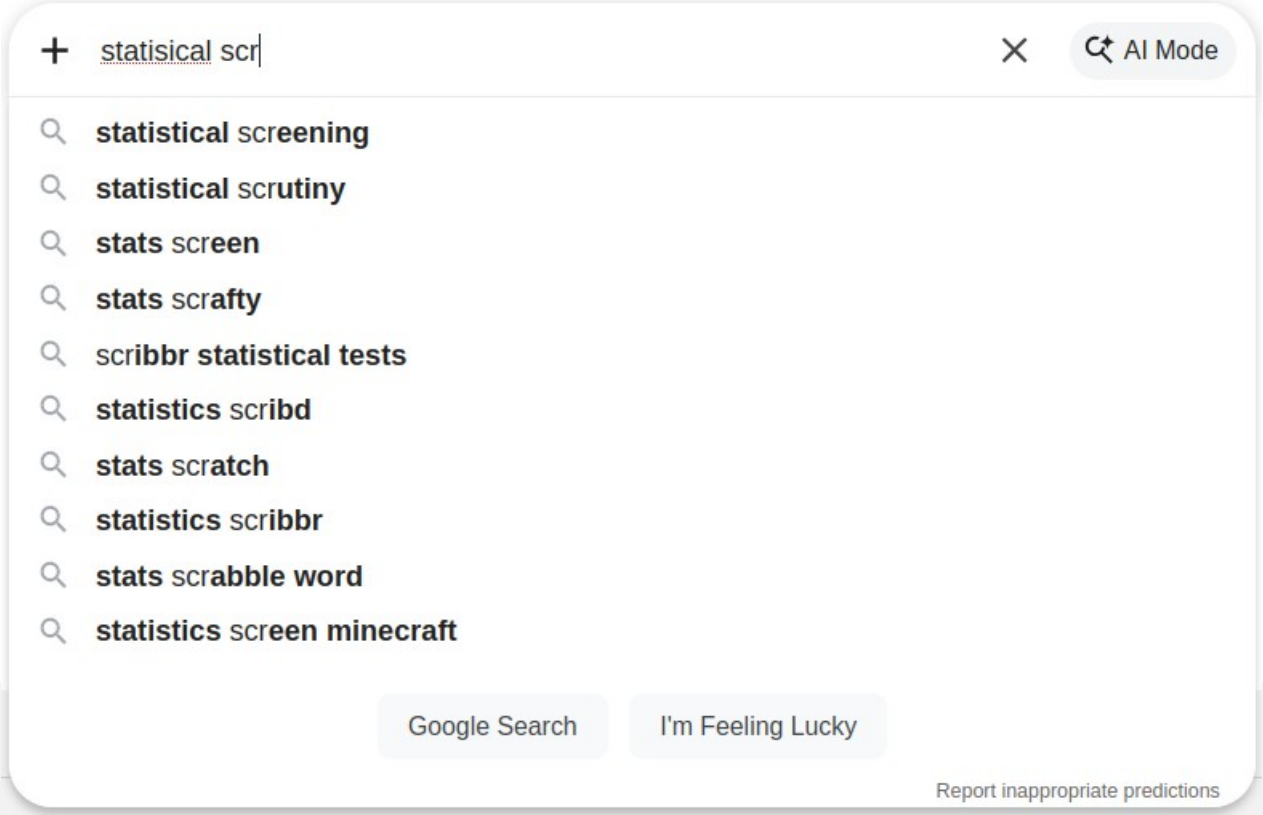
Exclude adult ideas Add filter

RESET

Keyword (by relevance)	Avg. monthly searches	Competition	Top of page bid (low range)	Top of page bid (high range)
protein powder	10K – 100K	High	£0.51	£2.58
whey protein	10K – 100K	High	£0.53	£2.11
protein shakes	100K – 1M	High	£1.61	£2.76
best protein powder	10K – 100K	High	£0.41	£1.57

SCRAPE FREQUENT QUERIES

https://google.com/



IDEA 2

1) Sample queries at random
   - Using query completions?
2) Assign a frequency to each query
   - Scraped from SEO site
3) Construct a stratified sample of queries
4) Scrape results too?
   - voting algorithm to merge results from multiple search engines

SCRAPE RESULTS

https://google.com/search/

Google

statistical scraping

AI Mode | All | Images | Videos | Forums | News | Short videos | More | Tools

Ssig | Example | Pdf

AI Overview

Statistical scraping, often called selective scraping, is **the automated extraction of specific, structured, or semi-structured data from websites to inform statistical analysis, research, or official statistics**. It replaces manual data collection with rapid, large-scale automated acquisition, commonly using tools like Python's BeautifulSoup and requests to populate databases or analysis tools like Pandas DataFrames. GitHub +3

Key Aspects of Statistical Scraping

- **Purpose:** Unlike general web scraping, statistical scraping aims to create high-quality datasets for modeling, index calculation (e.g., consumer prices), and market trends.

Show more

GitHub - SNStatComp/SSIG: Home of the Statistical Scraping Interest Group for methodologically sound use of web data i...

Statistical scraping is the use of web data in a way that the relationship with the statistical context is known. It often...

GitHub

Uses of web scraping for official statistics - CIRCABC

• one-time scraping for a research project to scraping for use. in production. • Web scrapin...

circabc.europa.eu

GitHub

https://github.com › SNStatComp › SSIG

SNStatComp/SSIG: Home of the Statistical Scraping ...

Roughly stated: **statistical scraping** is the use of web data in such way that the relationship with the statistical context is known. It often invloves search ...

SCRAPE RESULTS

https://bing.com/

statistical scraping

6

ALL | SEARCH | SHOPPING | IMAGES | VIDEOS | MAPS | COPILOT | MORE

About 56,700 results

Competify
nl.competify.nl › web-scraping

Website scraping - Alle data van elke website

Sponsored Ontvang nauwkeurige datasets en inzichten via API of ons gebruiksvriendelijke dashboard. Wij zetten elke website om in gestructureerde data en inzichten.

Uses of web scraping for official statistics

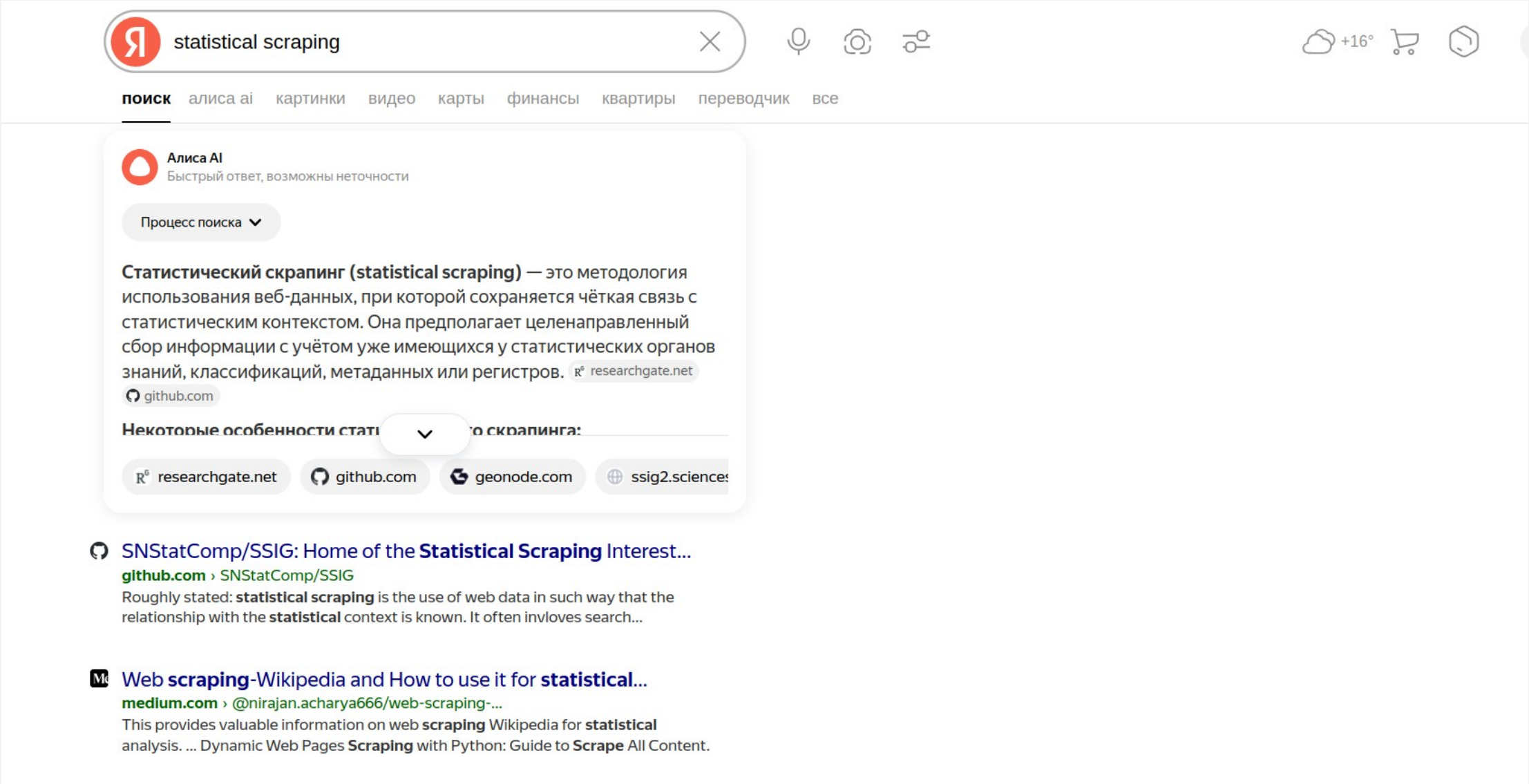
Web scraping is increasingly being utilized for official statistics to enhance the quality and accuracy of data collection. Here are some of the key uses of web scraping in this context:

- **Price Statistics**: Web scraping can be employed to gather price

Read more

SCRAPE RESULTS

https://yandex.ru/



SCRAPE RESULTS

https://baidu.com/

Bai du 百度

statistical scraping

百度一下

百度首页

文心 网页 图片 资讯 视频 笔记 地图 贴吧 文库 更多 搜索工具

百度Ai+ 听

“Statistical scraping” 通常指为统计分析目的而进行的数据抓取（web scraping），即从网页等公开来源自动提取数据，用于支持统计研究、政策制定或官方统计工作。

根据公开资料，以下是关键信息：

1. 定义与用途

- 网络数据抓取（Web Scraping）是指通过技术手段从网页中提取结构化或非结构化数据，并按规则处理后存入数据库的过程 12。
- 在统计领域，它被用于：
  - 弥补官方统计数据的时间滞后；
  - 构建新指标（如经济、社会、环境监测）；
  - 支持可持续发展目标（SDGs）等国际议程的数据需求 2 4。

2. 实际应用场景

- 官方统计机构（如德国联邦统计局）已试点将机器学习与数据抓取结合，用于商业统计、就业分析等 3。
- 国际组织（如联合国、IMF、世界银行）推动利用大数据和抓取技术提升国家统计能

展开

深度思考

12 Data Analytics Books for Beginners: A 2026 Reading List |...

相关搜索

- statistical
- statista数据库
- COMPUTATIONAL STATISTICS
- statistic软件
- the statistics proves
- stata分析

百度热搜 换一换

热搜榜 民生榜 财经榜

- 习近平会见西班牙首相桑切斯
- 1 中方从不承认所谓“阿鲁纳恰尔邦” 新
- 2 外卖员总结最不能点的外卖 热
- 3 从买商品到买“服务”
- 4 大批“五一”航班突然取消 热
- 5 外国人从中国买走50亿元保温杯
- 6 伊朗：若战争继续 将启用新军事能力 新
- 7 女子挖野菜挖到大蛇 双方都吓坏了
- 8 乌克兰对克里米亚发动大规模袭击 热

SCRAPE RESULTS

https://mojeek.com/

mojeek

statistical scraping

Mojeek User Survey

Web Summary Images News

Results 1 to 10 from 577,302 in 0.45s

https://www.clayford.net › statistics › scraping-data-off-a-web-site

Scraping Data off a Web Site – statistics you can

I could have scraped that as well but decided to enter ... Pingback: **Scraping** Virginia Tech Football Data, Part 2 | **Statistics** you can Probably Trust

https://blog.finxter.com › basketball-statistics-page-scraping-using-...

Basketball Statistics - Page Scraping Using Python and

The processes of this walkthrough can be easily applied to any number of their pages, but for this case, we plan on **scraping** seasonal **statistics** of ...

See more results from blog.finxter.com »

https://stackoverflow.com › questions › 54273839 › how-can-i-grab-data-fr...

web scraping - How can I grab data from this web site? - Stack

How accurate are the standard error formulas to find the standard deviation of the sampling distribution of a **statistic**?

https://stackoverflow.com › questions › 64232340 › using-beautifulsoup-to...

python - Using beautifulsoup to click through pages and then

From there, I'd like the **scraper** to click " **statistics** " and this will open up the URL that is needed to be scraped in a new window.

See more results from stackoverflow.com »

Summarise these results »

Su

HOME WORK

- N. Craswell and D. Hawking, Overview of the TREC-2004 Web Track, In: Proceedings of the 13th Text Retrieval Conference (TREC), 2004.
- D. Hawking, Overview of the TREC-9 Web Track, In: Proceedings of the 9th Text Retrieval Conference (TREC), 87–103, 2000.
- G. Pass, A. Chowdhury, and C. Torgeson. A Picture of Search. In: Proceedings of the 1st international conference on Scalable information systems (InfoScale), 1–7, 2006.
- W. Lai. Fitting power law distributions to data. UC Berkeley Statistics. 2016.
- S. Virkar. Power-law distributions and binned empirical data , 2012

Open WebSearch .eu